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c) BF_3 and NO_2-

d) BF_3 and NH_2^-

20. Some of the following properties of two species, NO_3^- and H_3O^+ are described below. which one of them is correct?

a) dissimilar in hybridisation for the central atom with different structure.

b) isostructural with same hybridisation for the Central atom.

c) different hybridiration for the central atom with same structure

d) none of these

21. The types of hybridiration on the five carbon atom from right to left in the, 2,3 pentadiene.

a) $sp^3, sp^2, sp, sp^2, sp^3$

b) sp^3, sp, sp, sp, sp^3

c) $sp^2, sp, sp^2, sp^2, sp^3$

d) $sp^3, sp^3, sp^2, sp^3, sp^3$

22. $Xe F_2$ is isostructural with

a) $SbCl_2$ b) $BaCl_2$

c) TeF_2 d) ICl_2^-

23. The percentage of s-character of the hybrid orbitals in methane, ethane, ethene and ethyne are respectively

a) 25, 25,33.3,50

b) 50,50,33.3,25

c) 50,25,33.3,50

d) 50,25,25,50

24. Of the following molecules, which have shape similar to carbondixide?

a) $SnCl_2$ b) NO_2

c) $C_2 H_2$ d) All of these.

25. According to VSEPR theory, the repulsion between different parts of electrons obey the order.

a) l.p – l.p > b.p–b.p> l.p–b.p

b) b.p–b.p> b.p–l.p> l.p–b.p

c) l.p–l.p> b.p–l.p > b.p–b.p

d) b.p–b.p> l.p–l.p> b.p–l.p

26. Shape of ClF_3 is

a) Planar triangular

b) Pyramidal

c) 'T' Shaped

d) none of these

27. Non- Zero dipole moment is shown by

a) CO_2

b) p-dichlorobenzene

c) carbontetrachloride

d) water.

28. Which of the following conditions is not correct for resonating structures?

a) the contributing structure must have the same number of unpaired electrons

b) the contributing structures should have similar energies

c) the resonance hybrid should have higher energy than any of the contributing structure.

d) none of these